MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 7MP7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb	Ehk	Rg1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	ft
Maximum:	6.9	0, -180 (See note 1)	770	8,800	±125	Note 1	60,000
Minimum:	5.7	---	---	4,000	---	---	---
TEST CONDITIONS:	6.3	Adjust	250	7,000	---	---	---

GENERAL:

Qualification - Required

Ⓖ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
5101	Face tilt	-		---	---	---	---
1141	Pressure (implosion)	-		---	---	---	---
5101	Neck and bulb alinement (magnetic types)	-		---	---	---	---
1002	Barometric pressure, reduced	-	Pressure = 64 mmHg	---	---	---	---
5111	Vibration	-		Width	---	2.0	mm
1331	Direct-interelectrode capacitance	-	k to all g1 to all	Ck Cg1	--- ---	8 10	pF pF
5201	Electrode current (grid No. 2)	-	Ec1 = 0	Ic2	---	15	µAdc
5216	Cathode illumination	-		---	---	---	---
(G) 1216	Base material insulating quality	-		---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown (magnetic types)	-		---	---	---	---
5206	Gas ratio	2		Gr	---	0.25	---
(G) 5223	Modulation	-	Ib = 200 µAdc	ΔEc1	---	38	Vdc
5101	Neck straightness	3		---	---	---	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5231	Spot position (magnetic deflection)	-		---	---	12	mm
5236	Zero-bias anode current (magnetic deflection)	-		---	---	---	---
5241	Grid-cutoff voltage	-		Ec1	-27	-63	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
---	Resolution at center	4	Ib = 200 µAdc	Res	350	---	Line
---	Aperture alinement	5		Distance	---	6.0	mm
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5216	Stray light emission (conventional types)	-	Eb = 8,800 Vdc; Ec2 = 770 Vdc	---	---	---	---
(G) 5101	Side terminal and base alinement	-	Vacant pin, position No. 3	---	---	---	---
5221	Screen	6		---	---	---	---
5251	Heater-cathode leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
---	Focusing coil current	7	Ib = 200 µAdc	I (coil)	72	98	mAdc

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
1101	Secureness of base, cap, or insert	-		---	---	---	---
(G) 1111	Base pin solder depth	-		---	---	---	---
1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb = 8,800 Vdc; Ec2 = 770 Vdc; Ib = 60 uAdc, t = 500 hours	---	---	---	---
---	Life-test end points:						
---	Center resolution	-		Res	300	---	Line
5223	Modulation	-		ΔEc1	---	38	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		---	---	---	---
5251	Heater-cathode leakage current	-		---	---	---	---
5216	Stray light emission	-		---	---	---	---

NOTES:

1. When Ec2 is greater than 330 Vdc, or Ec1 is near zero, or both, the effective resistance of the anode and grid 3 supply should be adequate to limit the anode and grid 3 input power to 6 watts. The peak grid 1 drive from cutoff should never exceed 65 volts.
2. This test to be performed at the conclusion of the holding period.
3. Neck and base straightness will be determined by the insertion of the tube neck in a cylinder 5 inches (127.00 mm) long and 1.500 $^{+0.003}_{-0.000}$ (38.10 $^{+0.08}_{-0.00}$ mm) inside diameter. This cylinder should move freely between the reference line and the base of the assembled tube.
4. Use raster scanning pattern 4 x 5-1/2 inches (101.60 x 139.70 mm) composed of 525 lines interlaced and synchronized. With Ec1 at cutoff vary signal from video generator utilizing a 2F21 monoscope type pattern (indian head) from zero to indicate Ib value. Focus for equal horizontal and vertical resolution in the center of the pattern.
5. The distance between the center of the unfocused, undeflected spot at low intensity (Ec1 near cutoff) and the center of the image of the masking aperture observed at high intensity of the unfocused, undeflected spot shall not exceed the limit specified. Ec1 should not be held at zero for more than approximately 10 seconds to prevent damage to the screen.
6. The test conditions shall be anode voltage (relative to cathode) 4,000 V minimum and beam current 60 µA maximum.
7. Read with JEDEC Standard Focus Coil No. 109, or equivalent. The distance from the center of the focus coil airgap to the reference line shall be 2-3/4 inches (69.85 mm) (see figure 1).

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Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
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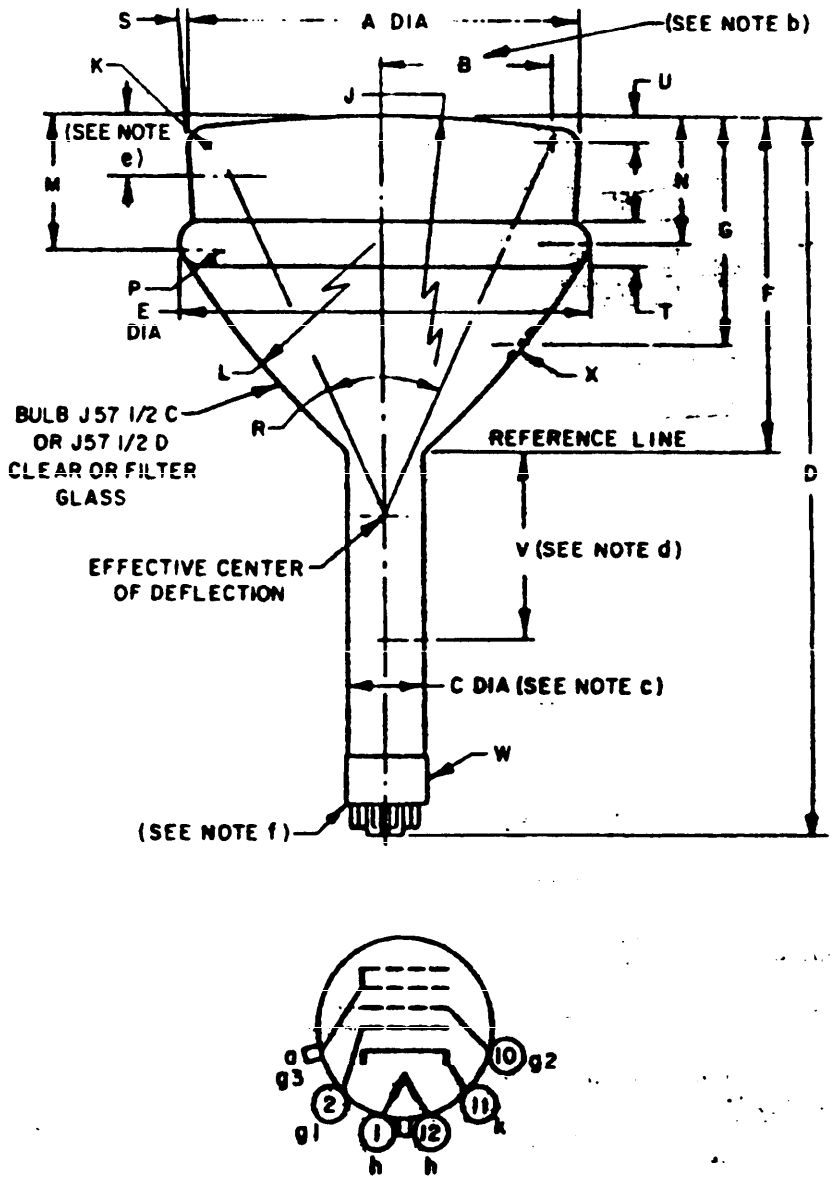


FIGURE 1. Outline drawing for electron tube type 7MP7.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Qualification inspection	
W	Base: B5-57	
X	Contact cap: J1-21	
	Quality conformance inspection, part 2	
A	7.062 (179.37)	7.312 (185.72)
B	3.000 (76.20) R	---
C	1.380 (35.05)	1.500 (38.10)
D	12.375 (314.32)	13.125 (333.38)
E	---	7.310 (185.67)
F	5.753 (146.13)	6.127 (155.63)
G	3.940 (100.08)	4.440 (112.78)
V	3.250 (82.55)	---
	Reference dimensions	
J	24.000 (609.60) R	
K	.380 (9.65) R	
L	15.000 (381.00) R	
M	2.400 (60.96)	
N	2.200 (55.88)	
P	.880 (22.35)	
R	50°	
S	3.5°	
T	.750 (19.05)	
U	.440 (11.18)	

NOTES:
a. Reference dimensions are for information only and are not required for inspection purposes.
b. Minimum useful screen radius.
c. Applies to entire length of neck below reference line. The reference line is determined by position where a gage 1.500 (36.10 mm) $^{+.003\ (.08\ mm)}_{-.000}$ inside diameter and 2 inches (50.80 mm) long will rest on bulb cone.
d. Deflection yoke shall be within this area.
e. The distance from 1 inch (25.40 mm) along tube wall may be used for support.
f. Bottom circumference of base shell will fall within circle with diameter of 1.875 (47.63 mm) and concentric with bulb axis.

FIGURE 1. Outline drawing for electron tube type 7MP7 - Continued.

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WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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